

TECHNICAL ARCHITECTURE OVERVIEW

The solution is constructed entirely within the enterprise ServiceNow platform ecosystem. The architecture utilizes native platform capabilities for data persistence, workflow automation, notifications, and security, avoiding third-party API dependencies.

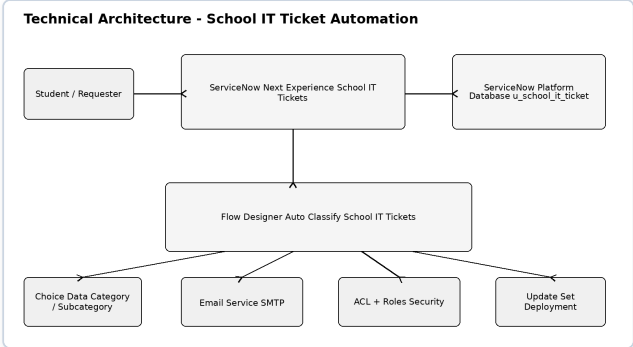


Figure 1: Technical architecture of the ServiceNow-based ticket automation solution.

TABLE 1: COMPONENTS & TECHNOLOGIES

S.No	System Component	Role & Description in Project	Technology Used
1	User Interface	Student ticket submission and technician queue management views.	ServiceNow Next Experience
2	Application Logic	Event-driven keyword parsing, classification, and notification flow.	ServiceNow Flow Designer
3	Data Store	Custom table storing tickets, choices, and assignment data.	u_school_it_ticket table
4	Cloud Database	Underlying relational database managed by ServiceNow Cloud.	ServiceNow Cloud DB
5	File Storage	Platform attachment engine (not required for prototype).	Not Used / Native
6	External API	No third-party REST/SOAP APIs needed; platform-native logic.	Not Used
7	Machine Learning Model	Rule/keyword based condition branches used for prototype.	Platform Flow Logic
8	Security Layer	Role-based Access Control Lists (ACLs) governing table access.	ServiceNow ACLs & Roles
9	Change Tracking	Captures schema, forms, flows, and choices for export.	ServiceNow Update Sets
10	Infrastructure	Enterprise cloud PaaS hosting the developer instance.	ServiceNow Cloud PaaS

TABLE 2: APPLICATION CHARACTERISTICS

S.No	Characteristic	Implementation Analysis	Technology Platform
1	Open-Source Frameworks	No external open-source application framework used; built natively on ServiceNow.	ServiceNow Native Platform
2	Security Implementations	Custom ticket table protected by read/write ACLs and authenticated user sessions.	ServiceNow ACLs & Roles
3	Scalable Architecture	Event-driven Flow Designer asynchronously processes tickets via the platform scheduler.	Flow Designer Engine
4	High Availability	Cloud PaaS infrastructure ensures continuous multi-tenant operational availability.	ServiceNow Cloud PaaS
5	Performance Optimization	Lightweight keyword matching executes in milliseconds upon ticket insertion.	Flow Trigger & Actions

IMPLEMENTATION EVIDENCE - AUTOMATION WORKFLOW

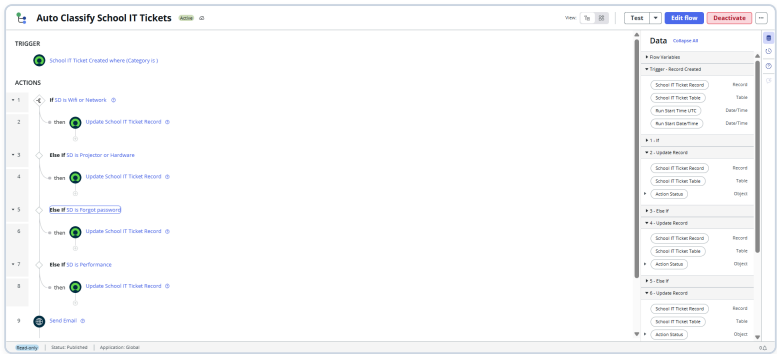


Figure 2: Published Flow Designer workflow implementing the core application logic.

IMPLEMENTATION EVIDENCE - DEPLOYMENT PACKAGE

The completed project configuration was captured into the official Update Set and exported as XML.

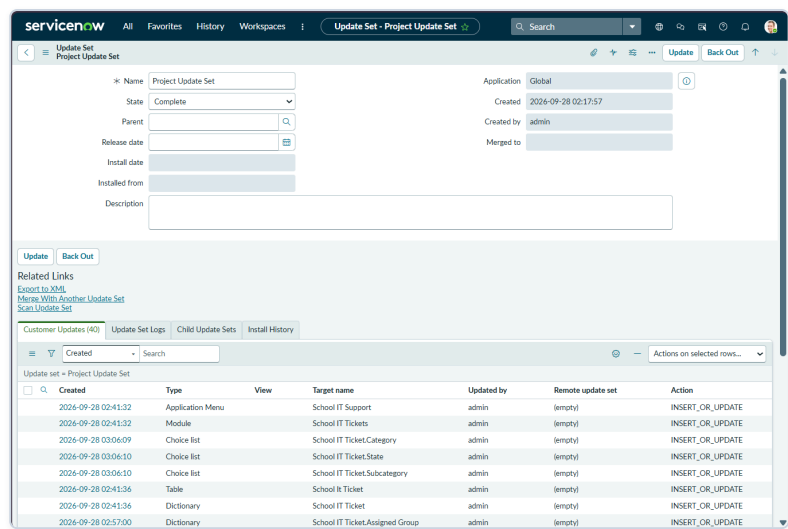


Figure 3: Completed Project Update Set with captured customer updates and XML export link.

Tech Stack Lead: Akash N A

Deployment Lead: Simeon P

Maximum Marks: 4 Marks